Inequalities — problem 7.81

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Prove that, for any positive numbers $x_1, x_2, \ldots, x_k$ $(k \geq 4)$,

$$\frac{x_1}{x_k + x_2} + \frac{x_2}{x_1 + x_3} + \cdots + \frac{x_k}{x_{k-1} + x_1} \geq 2.$$

Can you replace 2 by a greater number ?